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Cloud Computing

Amazon Web Services

BE CAREFUL!

Remember to delete every object you created in AWS when your project is corrected and the grade entered. This is to avoid AWS user fees.

REQUIRED TIME

You have 15 hours of class time to complete this project.

The project consists of two parts, each using AWS resources. The two parts are not connected so you can start with any of them.

Part 1: Saving student grades

Necessary AWS resources: Lambda, API Gateway, DynamoDB, S3 bucket.

Description:

You must create a static website that will show a form that lets users enter a student code, a midterm exam grade and a final exam grade. Once submitted, the form must pass its information to a **Lambda** function which will calculate the student's final grade using the following weighting: 40% for the midterm exam and 60% for the final exam. Once this calculation is done, a record containing the information must be written in a **DynamoDB** table named **StudentsGrades**.

If the student passes his course (60% or more), the record must contain four columns: **StudentCode**, **MidtermExam**, **FinalExam** and **FinalGrade**. If the student has failed, a fifth column must be added: **Comment** which will take the value of **"failed"**. Each student will only have one record.

Once your form is functional, you need to put it in an **S3** bucket and make it public as a static website.

Tips:

- Name your HTML file **index.html**
- Use the Lambda function **Node.js** course example as a reference.
- Don't forget to use an **API Gateway** to bridge the HTML form and the Lambda function.

Part 2: Display an instructor list

Necessary AWS resources: Amazon RDS, CloudFormation and EC2.

Description:

You need to code a small web application in PHP that will simply **list the content** of a MySQL database hosted on **Amazon RDS**. The database will be very simple and will only contain a table named **instructors** which has three columns: **InstructorCode**, **InstructorFirstname** and **InstructorLastname**.

Start by creating the PHP database and application locally. Once the application is functional, create a stack using the **Lamp Stack** AMI template (Linux, PHP and MySQL) in **CloudFormation**. This will create an **EC2** instance that contains a PHP and MySQL server.

However, you must not use the MySQL server of that instance. Instead, create a database in the **RDS** service and use that one with your PHP application. Make sure to enter a few instructors in the database. Name your PHP page **index.php** and drop the file on your **EC2** server in the appropriate place using **PuTTY** and the **pscp** command in order to make your site accessible on the internet.

WHAT TO HAND IN

Be sure to give the following documents to your instructor:

- Your AWS root username and password
- The URL link of your S3 bucket from part 1
- The URL link of your site hosted on EC2 in part 2
- The username and password for your instructor database in RDS

Correction scheme

You will be evaluated on the following points:

Project item	Points
• Lambda function calculates correctly and writes to database • Functional API Gateway • DynamoDB table created • S3 bucket created and exposes index.html file publicly • RDS database created and accessible • EC2 instance exists and exposes index.php file publicly • The index.php file displays the instructor list	25 15 10 10 10 15 15
Total	100

Penalties

- 5 % subtracted for each day late.